Tutorial – 05

Name : Ranudi Gayathmie Kariyapperuma
Index : 29897
Batch No : 23.1
Degree Program Name : Technology Management

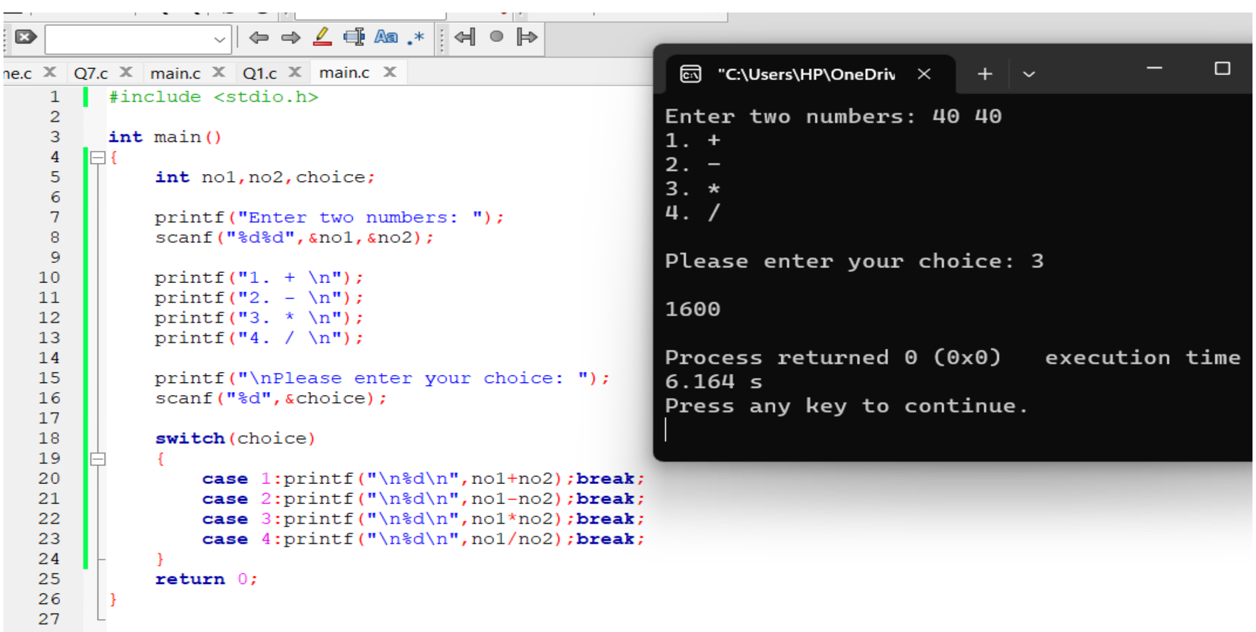
Switch

Input two numbers and display the outputs of the basic mathematic operations. The output screen should be displayed as follows;

Enter two numbers ____ ____

1. +
2. –
3. *
4. /

Please enter your Choice ___



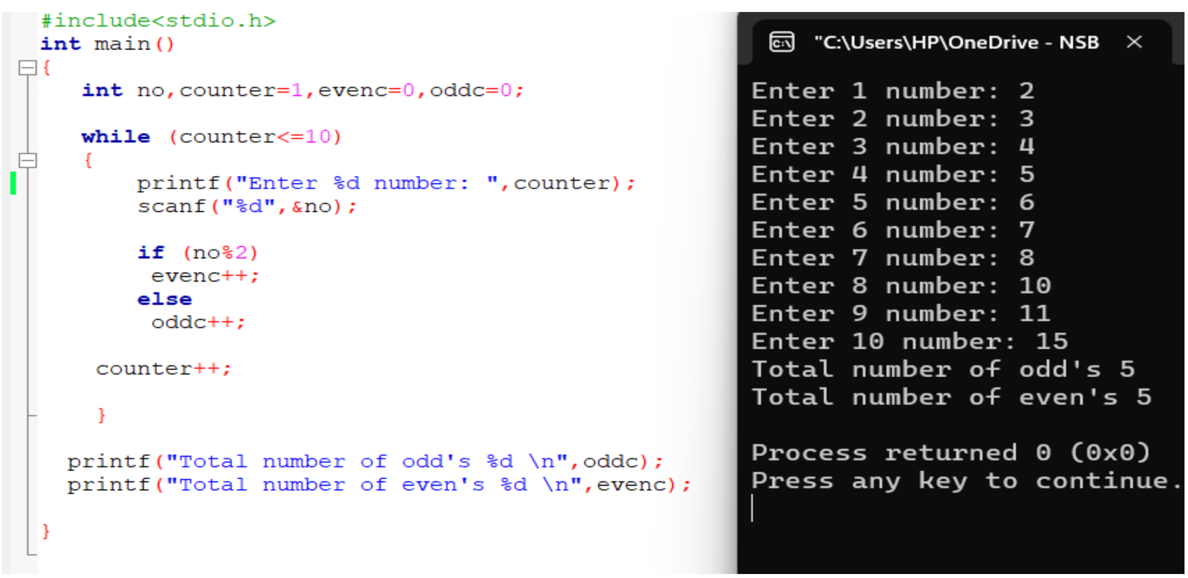
```c
#include <stdio.h>

int main()
{
    int no1,no2,choice;

    printf("Enter two numbers: ");
    scanf("%d%d",&no1,&no2);

    printf("1. + \n");
    printf("2. - \n");
    printf("3. * \n");
    printf("4. / \n");

    printf("\nPlease enter your choice: ");
    scanf("%d",&choice);

    switch(choice)
    {
        case 1:printf("\n%d\n",no1+no2);break;
        case 2:printf("\n%d\n",no1-no2);break;
        case 3:printf("\n%d\n",no1*no2);break;
        case 4:printf("\n%d\n",no1/no2);break;
    }
    return 0;
}
```

```
Enter two numbers: 40 40
1. +
2. -
3. *
4. /

Please enter your choice: 3

1600

Process returned 0 (0x0)   execution time
6.164 s
Press any key to continue.
```

While loop

1. **Input 10 numbers and display the total count of odd & even numbers in the entered number series.**

```
#include<stdio.h>
int main()
{
    int no,counter=1,evenc=0,oddc=0;
    while (counter<=10)
    {
        printf("Enter %d number: ",counter);
        scanf("%d",&no);
        if (no%2)
         evenc++;
        else
         oddc++;
         counter++;
     }
   printf("Total number of odd's %d \n",oddc);
   printf("Total number of even's %d \n",evenc);
}
```



2.Modify the above program in to enter series of numbers terminates when the user enter -99 and display the same expected output.

```
#include<stdio.h>
int main()
{
int no,evenc=0,oddc=0;
printf("Enter numbers (-99 to terminate)\n");
while (1)
{
printf("Number: ");
scanf("%d", &no);
if (no == -99)
{
break;
}
if (no%2)
evenc++;
else
oddc++;
}
printf("Total number of odd's %d \n",oddc);
printf("Total number of even's %d \n",evenc);

}
```

```
ooo.c  X  while2.c  X
```

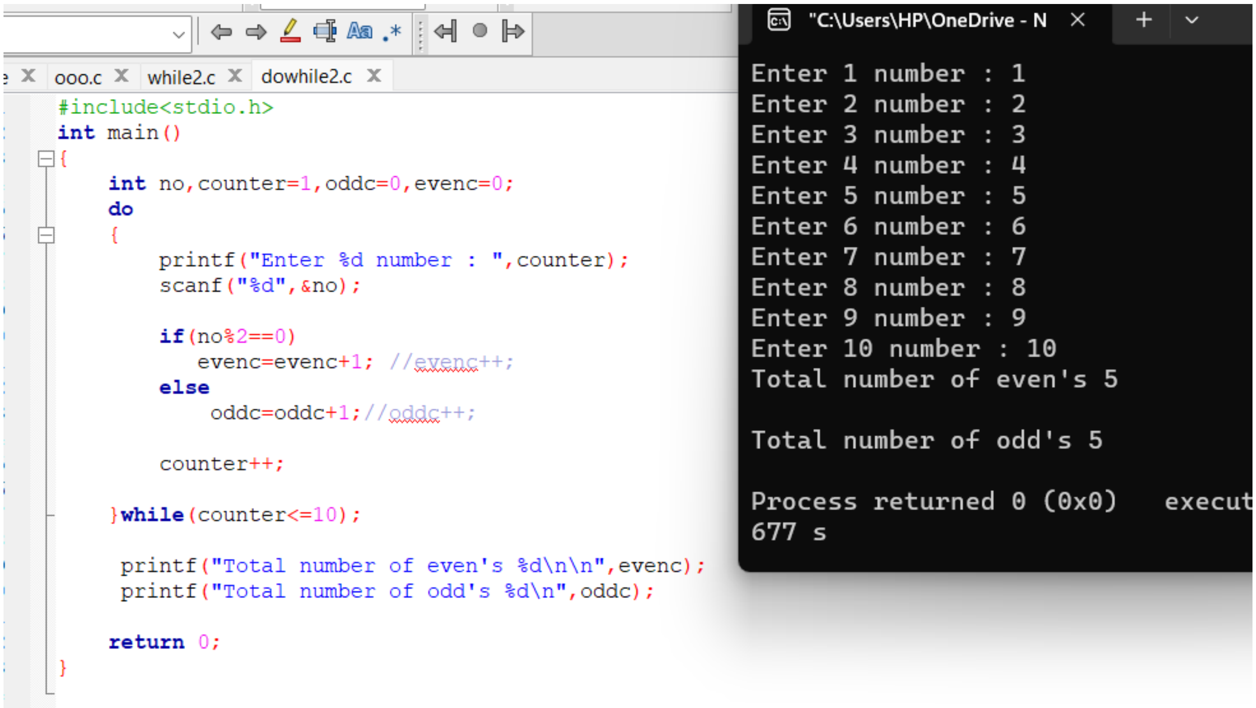
```c
#include<stdio.h>
int main()
{
    int no,evenc=0,oddc=0;

     printf("Enter numbers (-99 to terminate)\n");

    while (1)
    {
        printf("Number: ");
        scanf("%d", &no);

         if (no == -99)
         {
             break;
         }

        if (no%2)
         evenc++;
        else
         oddc++;
    }

  printf("Total number of odd's %d \n",oddc);
  printf("Total number of even's %d \n",evenc);

}
```

```
"C:\Users\HP\OneDriv  X  +

Enter numbers (-99 to terminate)
Number: 1
Number: 2
Number: 3
Number: 4
Number: 5
Number: 6
Number: 7
Number: 8
Number: 9
Number: 0
Number: 10
Number: 11
Number: 23
Number: 45
Number: -99
Total number of odd's 6
Total number of even's 8

Process returned 0 (0x0)   executio
18.189 s
Press any key to continue.
```

Do while loop

Rewrite the programs for the above while loop question 1 & 2 using do while loop

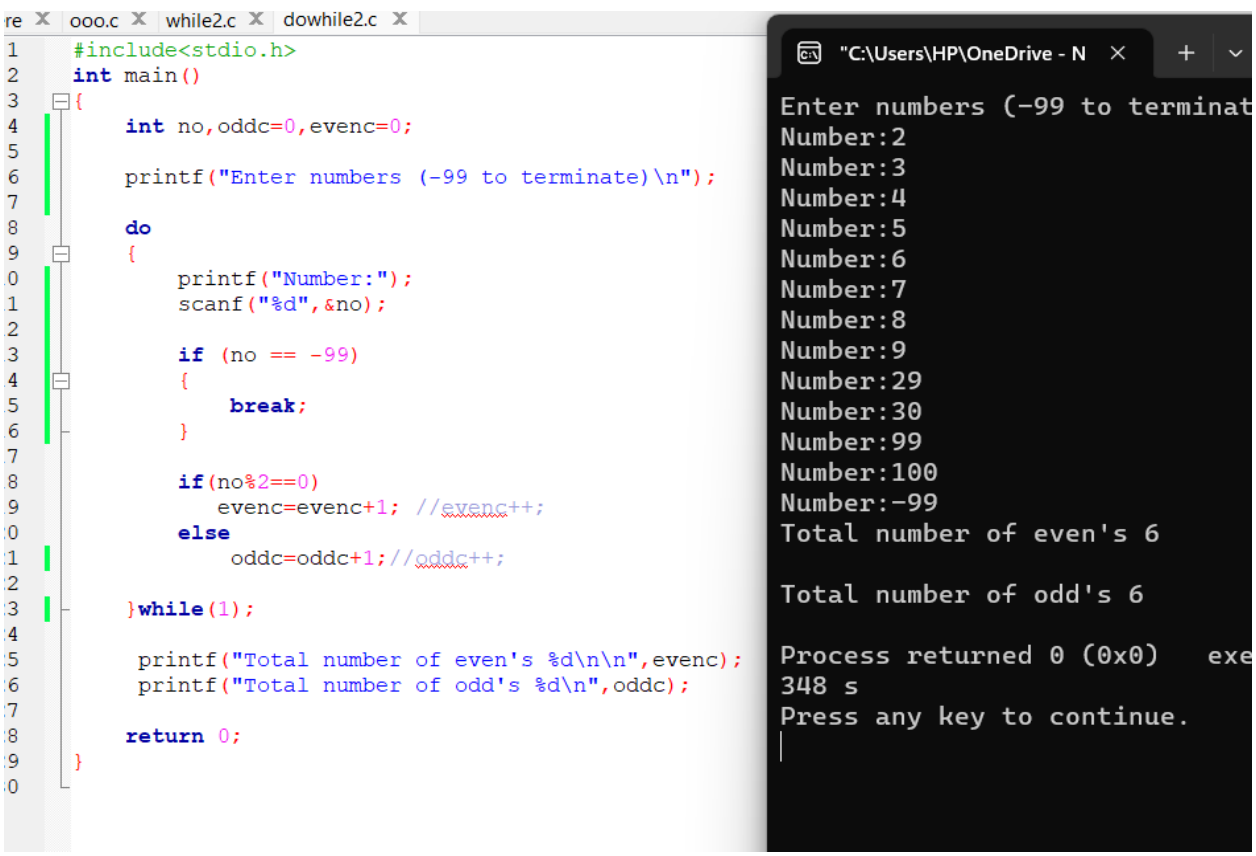
1
```
#include<stdio.h>
int main()
{
    int no,counter=1,oddc=0,evenc=0;
    do
    {
        printf("Enter %d number : ",counter);
        scanf("%d",&no);

        if(no%2==0)
           evenc=evenc+1; //evenc++;
        else
            oddc=oddc+1;//oddc++;

        counter++;

    }while(counter<=10);

     printf("Total number of even's %d\n\n",evenc);
     printf("Total number of odd's %d\n",oddc);

    return 0;
}
```



2.

```
#include<stdio.h>
int main()
{
    int no,oddc=0,evenc=0;

    printf("Enter numbers (-99 to terminate)\n");

    do
    {
        printf("Number:");
        scanf("%d",&no);

        if (no == -99)
        {
            break;
```

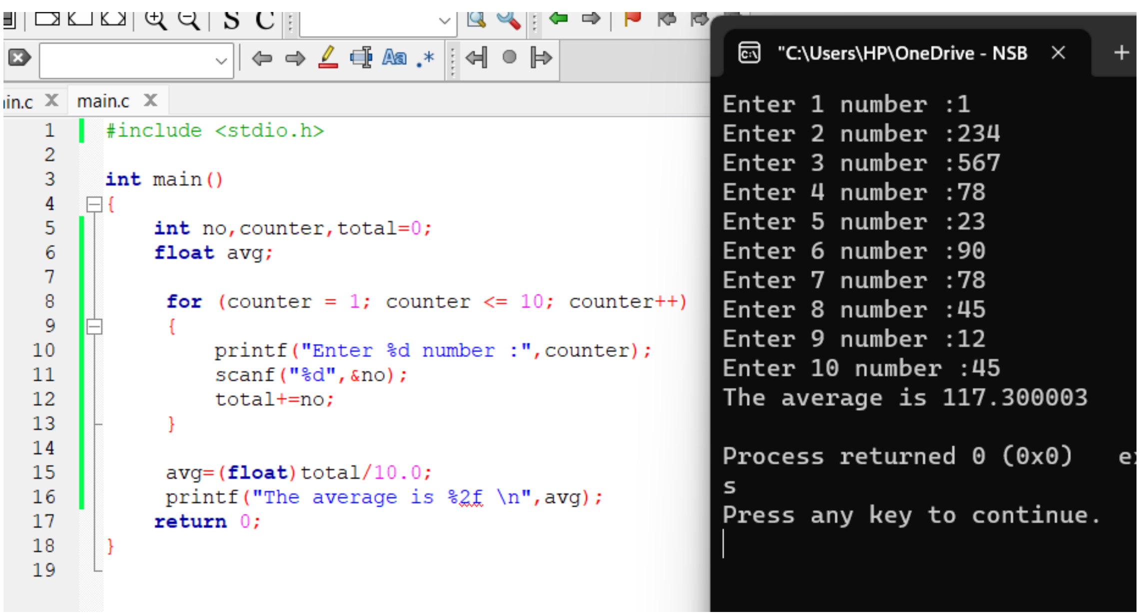
```c
        }

        if(no%2==0)
           evenc=evenc+1; //evenc++;
        else
            oddc=oddc+1;//oddc++;

    }while(1);

     printf("Total number of even's %d\n\n",evenc);
     printf("Total number of odd's %d\n",oddc);

    return 0;
}
```



For loop

1. **Input 10 numbers and display the average value using the for loop**

```
#include <stdio.h>
int main()
{
    int no,counter,total=0;
    float avg;
     for (counter = 1; counter <= 10; counter++)
     {
         printf("Enter %d number :",counter);
         scanf("%d",&no);
         total+=no;
     }
     avg=(float)total/10.0;
     printf("The average is %2f \n",avg);
    return 0;
}
```



2. Display the following output using the for loop

```
*

**

***

****

*****
```

```c
#include <stdio.h>
int main()
{
   int a,r;
   for(a=1;a<=5;a++)
   {
      for(r=1;r<=a;r++)
      {
         printf("*");
      }
         printf("\n");
   }
   return 0;
}
```